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Mark Newman

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CHAPTER

16 Epidemics on networks

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Abstract

This chapter discusses the spread of diseases over contact networks between individuals and the methods used to model this process. The chapter begins with an introduction to the classic models of mathematical epidemiology, including the SI model, the SIR model, and the SIS model. Models for coinfection and competition between diseases are also discussed, as well as "complex contagion" models used to represent the spread of information. The remainder of the chapter deals with the behavior of these models on networks, where the behavior of spreading diseases depends strongly on network structure. It is shown that the SIR model maps to a bond percolation process on networks, allowing us to solve for static properties such as the total number of individuals infected in a disease outbreak. The case of the configuration model is developed in detail and the calculations are extended to competing diseases, coinfection, and complex contagion. Time-dependent behavior of diseases on networks is also studied using various differential equation approximations, including pair approximations and degree-based approximations.

Keywords: Epidemiology, epidemic, disease, infection, SI model, SIR model, SIS model, coinfection, cross-immunity, complex contagion, percolation, edge percolation, bond percolation, pair approximation, moment closure

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